Types of Fossil Fuels

Chapter 2

Big Question

What are fossil fuels?

A train loaded with coal rolls toward a power plant. The train is powered by diesel fuel. The moving parts of the train are lubricated with different kinds of oil. As the engineer controls the train, he uses a plastic fork to eat lunch from a plastic container. Later that day, he goes home and uses a butane lighter to start a small coal fire in a stove to heat his home.

Every sentence in the paragraph above describes the use of at least one fossil fuel product. Coal, oil, and natural gas are three fossil fuels. People have developed many different substances from these fossil fuels, too.



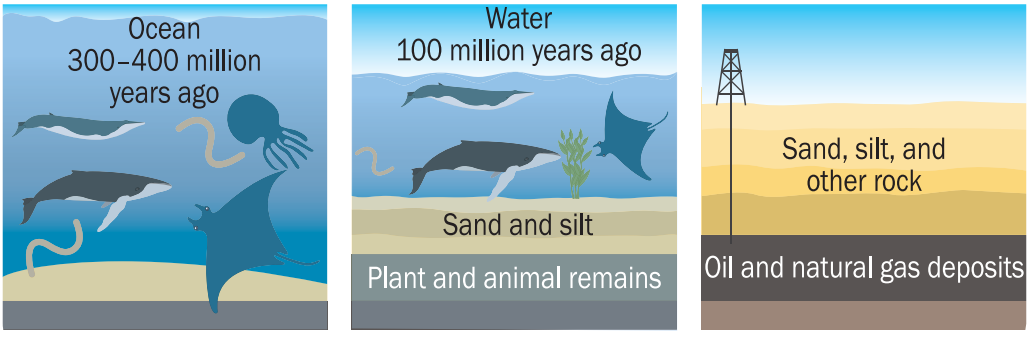
(Clockwise from top left) Petroleum jelly is made from crude oil extracted from Earth's crust. Kerosene is a fuel made from petroleum that can light lamps and heat stoves. Plastics like this fork and container are made from oil. Nylon fabric in backpacks is made from oil.

How Does Petroleum Form?

Fossil fuels are made of the remains of organisms. We call them fossil fuels because humans use them to fuel processes such as heating homes, powering vehicles, and producing electricity. All three fossil fuels are made of the remains of organisms that lived long ago.

Some marine organisms that died and settled on the ocean floor were then buried by layers of sand or mud. That sediment became rock over time. The weight of that rock on the organisms' remains provided pressure and intense heat that turned them into petroleum, or crude oil. Oil companies that look for and extract petroleum can study samples of the ocean floor. They use different technologies to identify areas that may hold oil deep under the surface. This work costs a lot of money. But there is a lot of money to be made back, because oil is widely used by people for many purposes.

Petroleum and Natural Gas Formation



Ocean organisms die. They become buried by silt and sand on the ocean floor.

The remains are buried deeper over millions of years. Heat and pressure turn the material into oil.

Petroleum deposits are reached by drilling through newer layers of rock.

How Do Coal and Natural Gas Form?

Coal forms in a similar way. Millions of years ago, Earth's landmasses were covered with plants. Shallow seas rose and fell, killing many of the plants. Some plants fell into swamps when they died. Their remains were buried under layers of mud. There was not enough oxygen available in the mud to allow the remains to decompose. Instead, they became a substance called peat. As more sediment built up over the peat, the peat pressed together and became solid. This is coal. Coal is very abundant and has been found all over Earth.

Natural gas forms where petroleum and coal form. Because gas is less dense than solids and liquids, it rises in Earth's crust until it gets stuck under a layer of rock. Sometimes natural gas is pumped out of an area above an oil field before the oil is removed. Methane is the most abundant gas in the mixture.



This exposed rock on an Australian beach shows a distinct layer of coal. Pressure from the layers of rock above turned soft peat into this black solid.

Varieties of Coal, Petroleum, and Natural Gas

There are different forms of coal, petroleum, and natural gas. Coal that has been under the greatest pressure is harder, blacker, and shinier than other types. The lightest coal tends to be brown and is considered low grade. In between are forms that are black but fairly dull and not as dense as the shiny, high-grade coal.

Petroleum is a liquid in Earth's crust. The liquid can be very thin and light to thick and heavy. This means that some crude oil floats if spilled at sea and some sinks to the ocean floor.

Natural gas consists mostly of methane, but it can also contain propane and butane in gas form. Methane is piped into homes and businesses for heating and cooking. Propane is popular for use in small tanks for barbecues. Butane is often used in lighters.



What begins as peat on the surface can eventually become coal. The deeper it is, the more pressure and heat it encounters. Different types of coal result from different amounts of pressure and heat.